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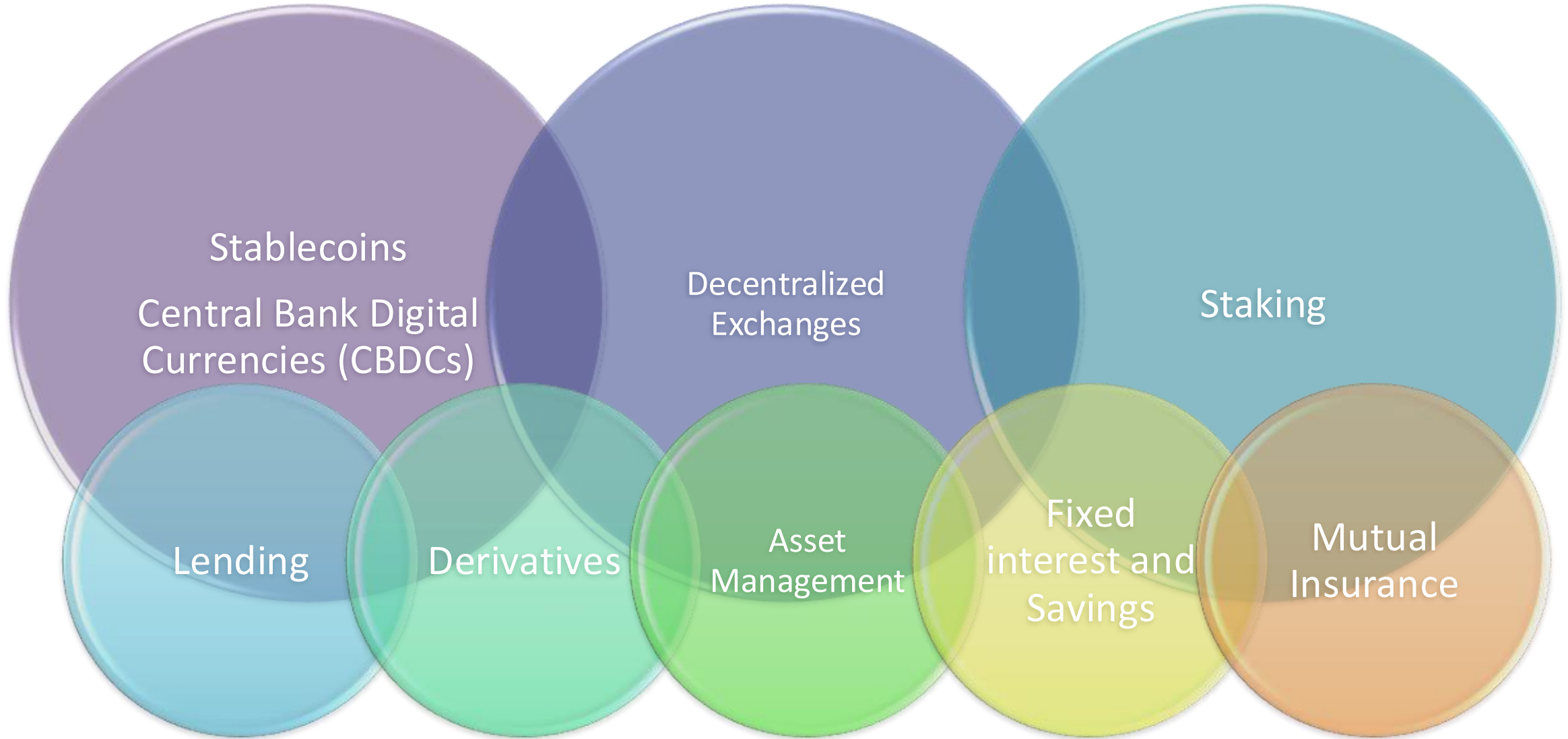
Lecture 3: DeFi 2

DeFi use cases

Lending, Exchanges, Savings, Insurance

SC6125 – Blockchain in Finance

Categories of DeFi



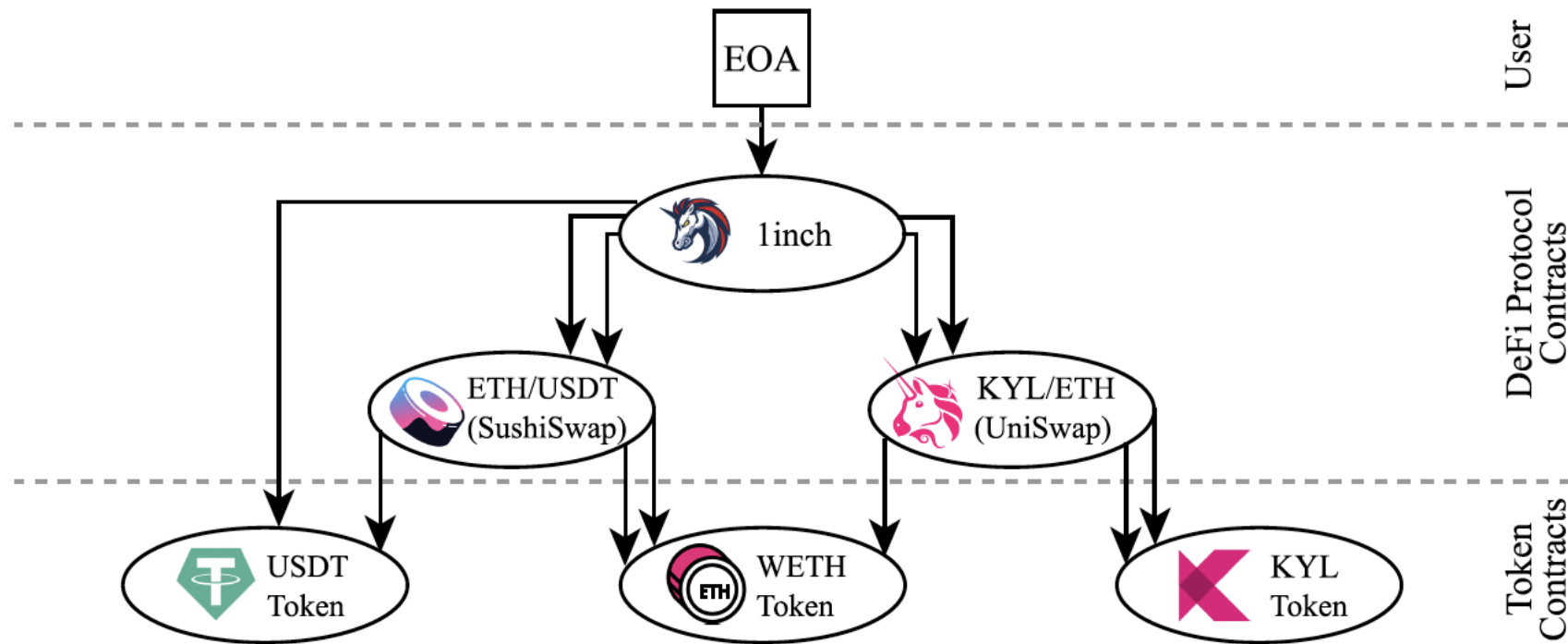
DeFi Compositions

- The term “financial lego” is sometimes used because DeFi services can be composed into new financial product
- A smart contract must always be initially called by an external transaction originating from an externally owned account
- Smart Contracts can call each other (using internal transactions or messages), and their individual functions can be arbitrarily composed into new financial products and services

Kitzler, S., Victor, F., Saggese, P., & Haslhofer, B. (2023). Disentangling decentralized finance (DeFi) compositions. *ACM Transactions on the Web*, 17(2), 1-26.



DeFi Compositions



- A DeFi composition in which USDT tokens are swapped against KYL tokens through the DeFi service 1inch in a single transaction
- 1inch executes the swap sequentially through the DeFi services SushiSwap and UniSwap, using WETH as an intermediary token

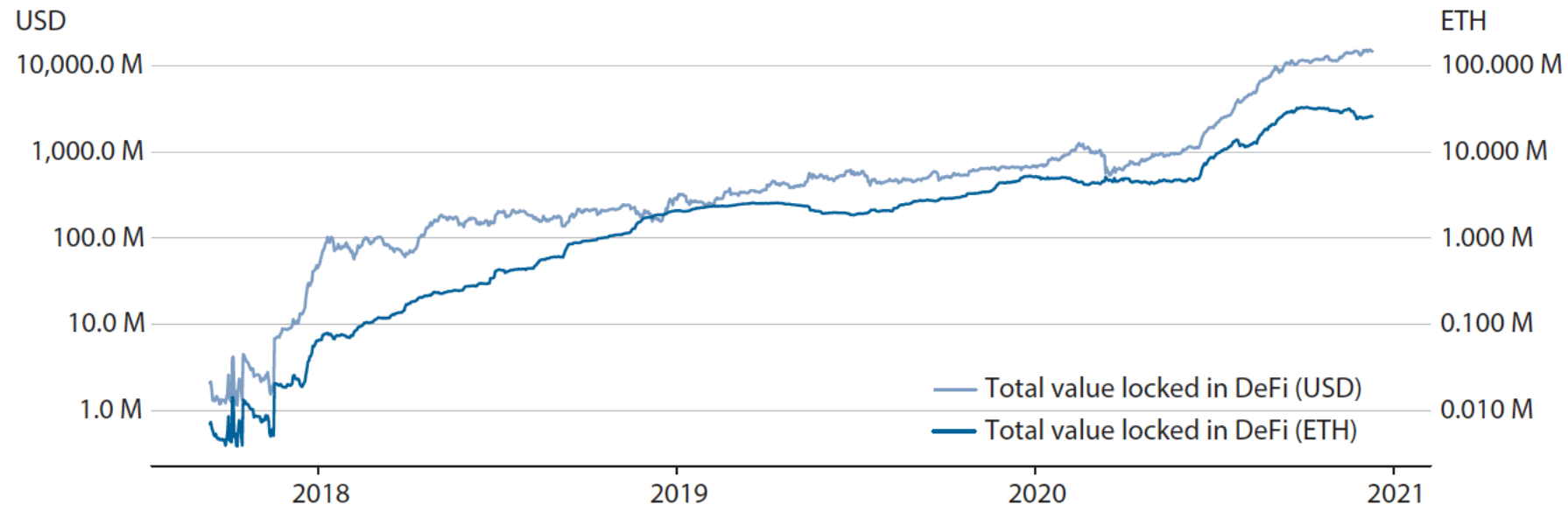
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Decentralized Finance

The value of funds that are locked in DeFi-related smart contracts crossed 10 billion USD in 2021

Figure 1

Total Value Locked in DeFi Contracts (USD and ETH)



NOTE: M, million.

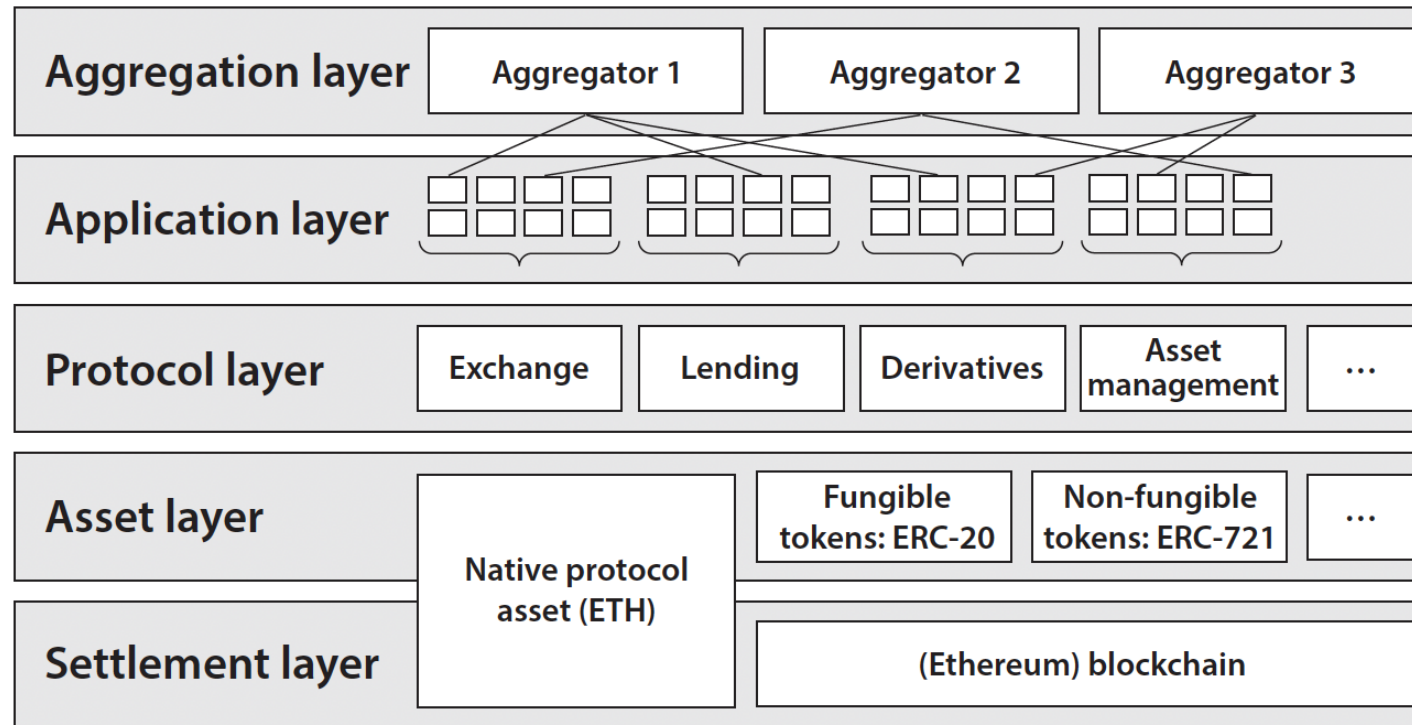
SOURCE: DeFi Pulse.

Schär, F. (2021). Decentralized finance: On blockchain-and smart contract-based financial markets. FRB of St. Louis Review.

Decentralized Finance

Figure 2

The DeFi Stack



Schär, F. (2021). Decentralized finance: On blockchain-and smart contract-based financial markets. FRB of St. Louis Review.

Decentralized Finance

- The settlement layer (Layer 1)
 - Consists of the blockchain and its native protocol asset (e.g., Bitcoin [BTC] on the Bitcoin blockchain and ETH on the Ethereum blockchain)
 - It allows the network to store ownership information securely and ensures that any state changes adhere to its ruleset
 - The blockchain can be seen as the foundation for trustless execution and serves as a settlement and dispute resolution layer

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Decentralized Finance

- The asset layer (Layer 2)
 - Consists of all assets that are issued on top of the settlement layer
 - This includes the native protocol asset as well as any additional assets that are issued on this blockchain (usually referred to as tokens)
- The protocol layer (Layer 3)
 - Provides standards for specific use cases such as decentralized exchanges, debt markets, derivatives, and on-chain asset management
 - These standards are usually implemented as a set of smart contracts and can be accessed by any user (or DeFi application)
 - As such, these protocols are highly interoperable

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Decentralized Finance

- The application layer (Layer 4)
 - Creates user-oriented applications that connect to individual protocols
 - The smart contract interaction is usually abstracted by a web browser-based front end, making the protocols easier to use
- The aggregation layer (Layer 5)
 - An extension of the application layer
 - Aggregators create user-centric platforms that connect to several applications and protocols
 - They usually provide tools to compare and rate services, allow users to perform otherwise complex tasks by connecting to several protocols simultaneously, and combine relevant information in a clear and concise manner

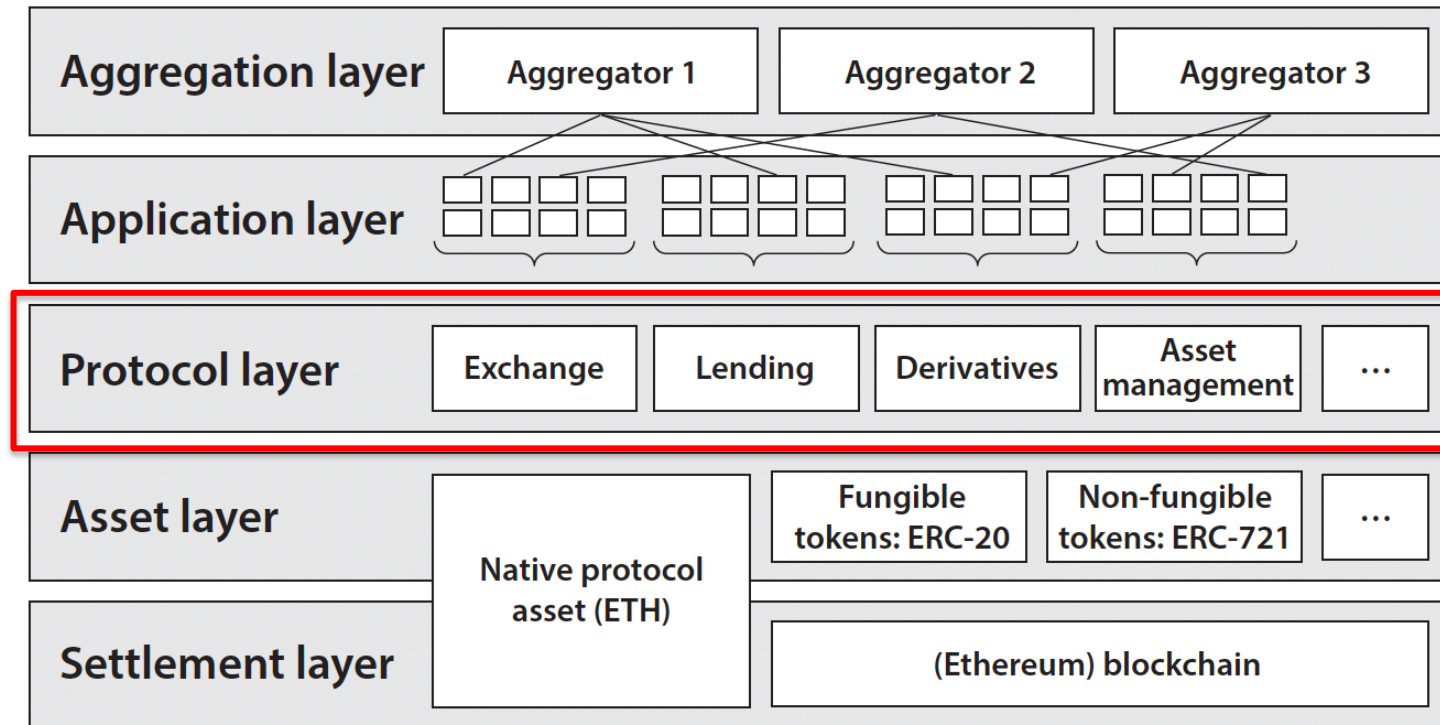
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Decentralized Finance

Figure 2

The DeFi Stack



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Decentralized Finance – Decentralized Exchange Protocol

- Centralized exchanges are relatively efficient, but they have one severe problem
 - To be able to trade on a centralized exchange, traders must first deposit assets with the exchange
 - They thereby forfeit direct access to their assets and have to trust the exchange operator
- Decentralized exchange protocols try to mitigate these issues by removing the trust requirement
 - Users no longer must deposit their funds with a centralized exchange
 - Instead, they remain in exclusive control of their assets until the trade is executed
 - Trade execution happens atomically through a smart contract, meaning that both sides of the trade are performed in one indivisible transaction, mitigating the counterparty credit risk

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Decentralized Finance – Decentralized Exchange Protocol

Decentralized Order Book Exchanges

- **On-chain order books** have the advantage of being entirely decentralized
 - Every order is stored within the smart contract
 - But it is a costly and slow process for which even the declaration of the intent to trade results in network fees
- **Off-chain order books** are hosted and updated by centralized third parties, usually referred to as relayers
 - They provide takers with the information they need to select an order they would like to match
 - Relayers are never in control of the funds and neither match nor execute the orders
 - The openness of the protocol ensures that there is competition among the relayers and mitigates potential dependencies
 - Eg, 0x

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Decentralized Finance – Decentralized Exchange Protocol

Constant Function Market Maker (or AMM)

- A constant function market maker (CFMM) is a smart contract-liquidity pool that holds (at least) two cryptoassets in reserve and allows anyone to deposit tokens of one type and thereby to withdraw tokens of the other type
- To determine the exchange rate, smart contract-based liquidity pools use variations of the constant product model, where the relative price is a function of the smart contract's token reserve ratio

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Decentralized Finance – Decentralized Exchange Protocol

Constant Function Market Maker or (AMM)

- It is important to point out that smart contract-based liquidity pools are not reliant on external price feeds (so-called oracles)
 - Whenever the market price of an asset shifts, anyone can use the arbitrage opportunity and trade tokens with the smart contract until the liquidity pool price converges to the current market price
 - The implicit bid/ask spread of the constant product model (plus a small trading fee) may lead to the accumulation of additional funds
 - Anyone who provides liquidity to the pool receives pool share tokens that allow them to participate in this accumulation and to redeem these tokens for their share of a potentially growing liquidity pool
- Prominent examples of smart contract-based liquidity pool protocols are UniSwap, Balancer, Curve, and Bancor

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Automated Market Maker

- AMMs using a constant product function solve the problem of on-chain asset exchange without centralized order books or market makers. The objective is to enable:
 - Continuous liquidity
 - Deterministic pricing
 - Permissionless trading
 - Atomic settlement
- Formally, the AMM maintains a pricing invariant:
 - $x \cdot y = k$
- where:
 - x = reserve of token A
 - y = reserve of token B
 - k = constant

Automated Market Maker

- In traditional markets:
 - Liquidity provisioning is centralized (market makers, exchanges)
 - Order matching requires trusted operators
 - Custody relies on exchange solvency
 - Execution fairness relies on opaque matching engines
- Centralized exchanges can technically solve trading efficiency but cannot provide:
 - Trust-minimized execution
 - Verifiable reserve backing
 - Permissionless access
 - Atomic, self-custodial settlement
- The problem is not technical matching, but credibly neutral execution without trust in an intermediary

Automated Market Maker

On-chain components

- Liquidity Pool Contract
 - Holds token reserves
 - Enforces $x \cdot y = k$
 - Executes swaps
 - Mints/burns LP tokens
 - Collects protocol fee
- Router Contract
 - User-facing abstraction
 - Handles multi-hop swaps
 - Computes quotes
 - Wraps ETH, handles approvals
- LP Token Contract
 - ERC-20 representing pool shares
 - Pro-rata claim on reserves
- Governance / control contracts (optional)
 - Parameter management (fees)
 - Emergency pauses
 - Protocol-owned liquidity controls



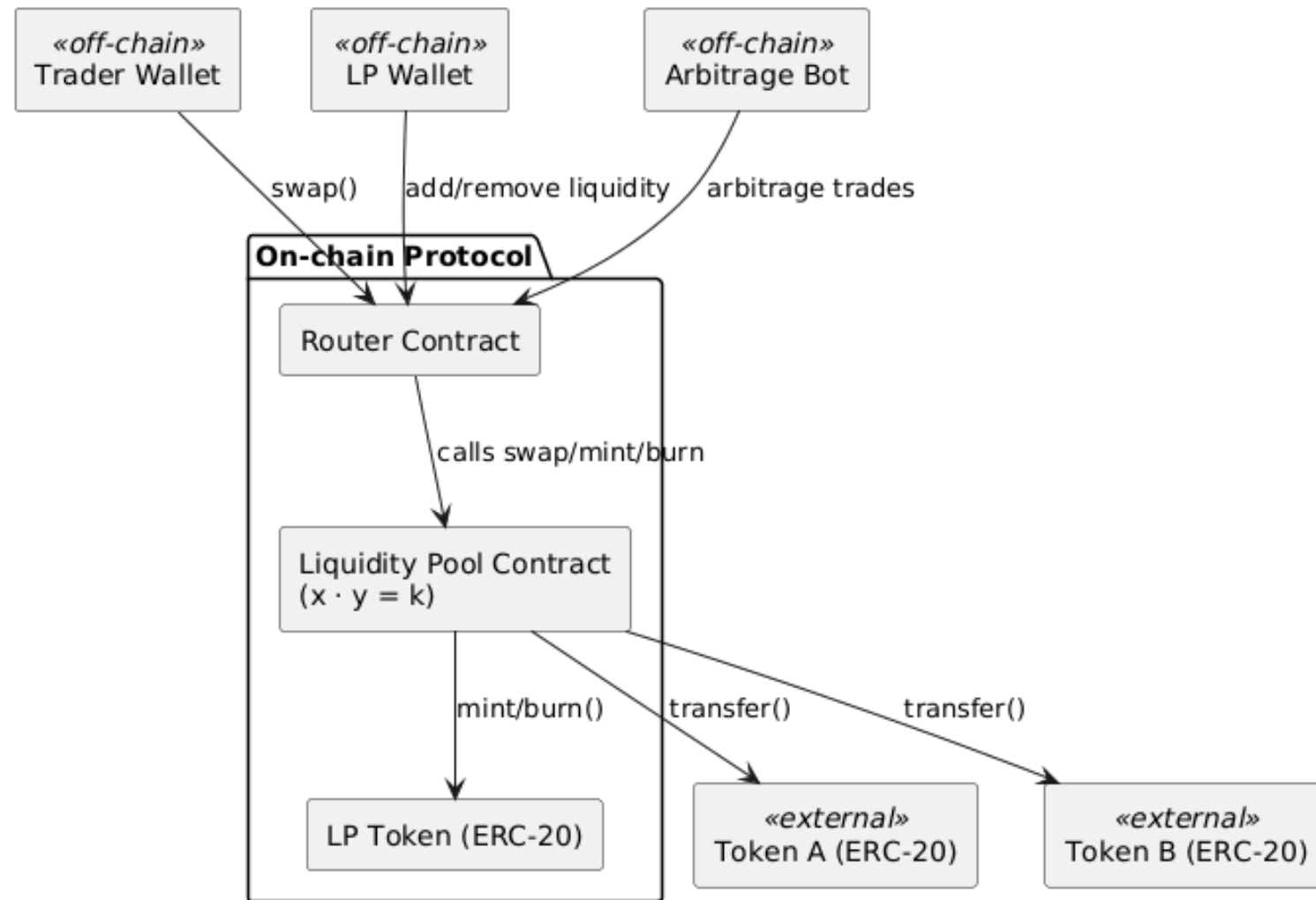
Automated Market Maker

Off-chain components

- User Interfaces
 - DEX front-ends
 - Wallets
- Bots / Agents
 - Arbitrage bots
 - MEV searchers
- External Dependencies
 - No oracle needed for pricing
 - Oracles may be used for UI display only



Automated Market Maker



Automated Market Maker

- Slippage is the difference between:
 - The expected execution price based on current pool reserves
 - The actual execution price after the trade moves the invariant
- In a constant product AMM:
 - Larger trades cause non-linear price movement
 - Slippage increases with trade size relative to pool depth
 - Slippage is structural, not accidental
- Mathematically, price is:
 - $P = dy/dx$
- and changes continuously during execution

Automated Market Maker

Token Swap with Slippage

- Initial pool state
 - ETH reserve (x) = 100
 - USDC reserve (y) = 200,000
 - Implied spot price = 2,000 USDC / ETH
 - $k = 20,000,000$
- User swaps 1 ETH $\rightarrow$ USDC

Naive expectation (no slippage):

$$1 \times 2,000 = 2,000 \text{ USDC}$$

Actual AMM execution:

$$(100 + 1)(y - \Delta y) = 20,000,000$$

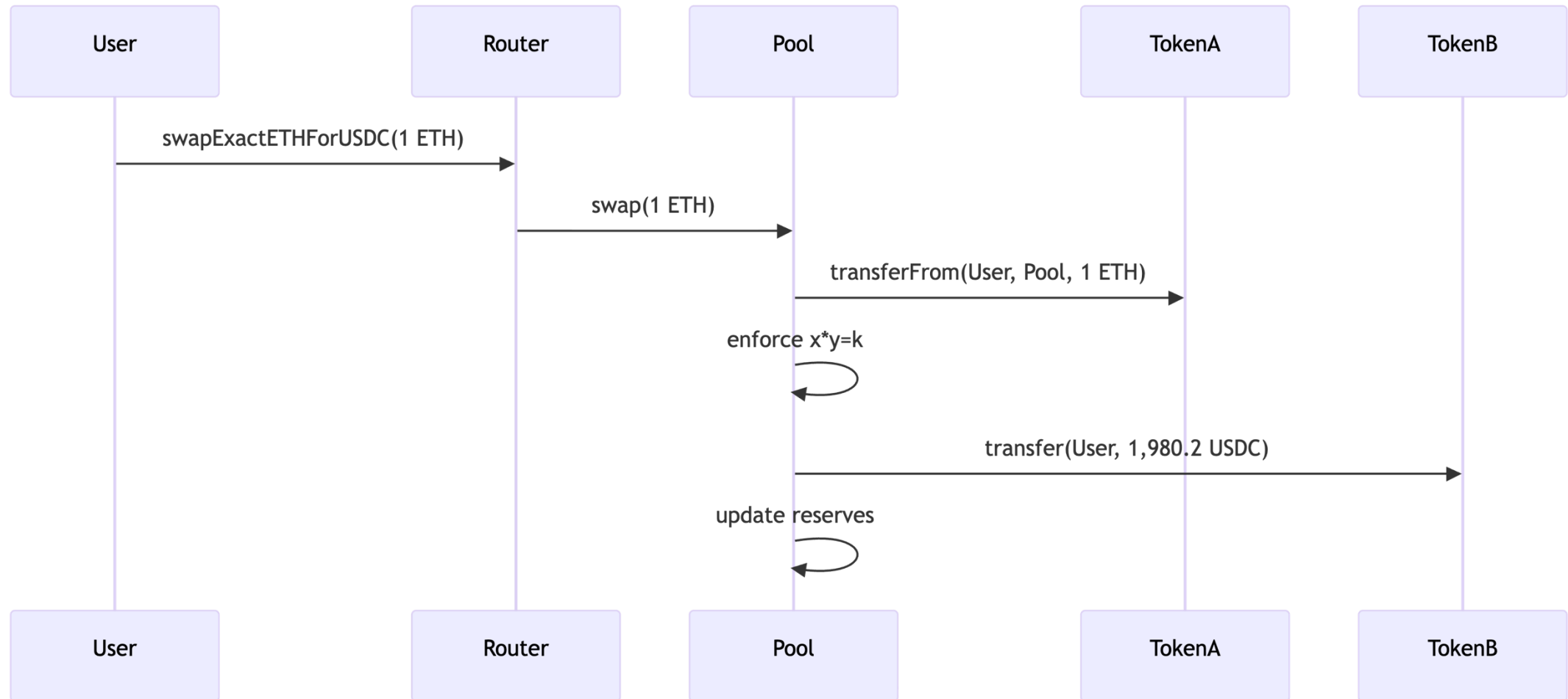
$$101 \cdot y' = 20,000,000 \Rightarrow y' \approx 198,019.8$$

USDC received:

$$200,000 - 198,019.8 = 1,980.2$$

$$\text{Slippage} = \frac{2,000 - 1,980.2}{2,000} \approx 0.99\%$$

Automated Market Maker



Automated Market Maker

- What if 30 ETH is swapped instead?
 - Naïve expectation = $30 \times 2,000 = 60,000$ USDC

New ETH reserve after trade

$$x_1 = x_0 + 30 = 130$$

The invariant must still hold:

$$x_1 \cdot y_1 = k$$

$$130 \cdot y_1 = 20,000,000$$

$$y_1 = \frac{20,000,000}{130} \approx 153,846.15$$

Absolute slippage

$$\text{Slippage (USDC)} = 60,000 - 46,153.85 = 13,846.15$$

Percentage slippage

$$\text{Slippage \%} = \frac{13,846.15}{60,000} \approx 23.1\%$$

Automated Market Maker

Incentives

- LPs earn fees, compensated for slippage risk
- Traders pay slippage implicitly
- Arbitrageurs align on-chain and off-chain prices
- Protocol may collect a fee share

Governance

- Sets fee parameters
- Enables protocol fee switch
- May pause system in emergencies

Automated Market Maker

Benefits

- Fully autonomous liquidity
- No order book management
- Universal accessibility
- Composable building block

Trade-offs

- Slippage increases non-linearly
- Capital inefficiency for volatile pairs
- Passive LP exposure to price movement
- Gas costs per interaction

Automated Market Maker

- Functions used in AMMs

Function	Formula	Key Difference
Constant Product	$x \cdot y = k$	General-purpose, inefficient
Constant Sum	$x + y = k$	Zero slippage, breaks with volatility
StableSwap (Curve)	Hybrid	Capital-efficient near peg
Concentrated Liquidity (Uniswap v3)	Piecewise CP	Capital efficient, more complex
CFMM with Oracles	Variable	External trust introduced

Group Discussion

- Research and discuss how these AMM functions deal with slippage
 - StableSwap
 - Uniswap v3

Decentralized Finance – Decentralized Exchange Protocol

Smart Contract-Based Reserve Aggregation

- Another approach is to consolidate liquidity reserves through a smart contract that allows large liquidity providers to connect and advertise prices for specific trade pairs
- A user who wants to exchange token x for token y may send a trade request to the smart contract
- The smart contract will compare prices from all liquidity providers, accept the best offer on behalf of the user, and execute the trade
- It acts as a gateway between users and liquidity providers, ensuring best execution and atomic settlement

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Decentralized Finance – Decentralized Exchange Protocol

Smart Contract-Based Reserve Aggregation

- Prices are set by the liquidity providers
 - This approach works fine if there is a relatively broad base of liquidity providers
 - However, if there is limited or no competition for a given trade pair, the approach may result in collusion risks or even monopolistic price setting
 - As a countermeasure, reserve aggregation protocols usually have some (centralized) control mechanisms, such as maximum prices or a minimum number of liquidity providers
- The best-known implementation of this concept is the Kyber Network

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Decentralized Finance – Decentralized Exchange Protocol

Peer-to-Peer Protocols

- An alternative to classic exchange or liquidity pool models are peer-to-peer (P2P) protocols, also called over-the-counter (OTC) protocols
- In contrast to other protocols, offers can be accepted exclusively by the parties who have been involved in the negotiation
- To make things more efficient, the process is usually automated
- AirSwap is the most popular implementation of a decentralized P2P protocol

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Decentralized Finance - Decentralized Lending Platforms

- There are a large variety of protocols that allow people to lend and borrow cryptoassets
 - Decentralized loan platforms are unique in the sense that they require neither the borrower nor the lender to identify themselves
 - Everyone has access to the platform and can potentially borrow money or provide liquidity to earn interest
 - As such, DeFi loans are completely permissionless and not reliant on trusted relationships
- To protect the lender and stop the borrower from running away with the funds, there are two distinct approaches:
 - Collateralized Debt Positions
 - Collateralized Debt Markets

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Decentralized Finance - Decentralized Lending Platforms

Collateralized Debt Positions

- Some DeFi applications allow users to create collateralized debt positions and thereby issue new tokens that are backed by the collateral
- To be able to create these tokens, the person must lock cryptoassets in a smart contract
- The number of tokens that can be created depends on the target price of the tokens generated, the value of the cryptoassets that are being used as collateral, and the target collateralization ratio

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Decentralized Finance - Decentralized Lending Platforms

Collateralized Debt Positions

- The newly created tokens are essentially fully collateralized loans that do not require a counterparty and allow the user to get a liquid asset while maintaining market exposure through the collateral
- The loan can be used for consumption, allowing the person to overcome a temporary liquidity squeeze or to acquire additional cryptoassets for leveraged exposure
- MakerDAO uses this method, MakerDAO has recently switched to a multi-collateral system, with the goal to make the protocol more scalable by allowing a variety of cryptoassets to be used as collateral

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Decentralized Finance - Decentralized Lending Platforms

Collateralized Debt Markets

- Instead of creating new tokens, it is also possible to borrow existing cryptoassets from someone else
- In other words: For someone to be able to borrow ETH, there must be another person willing to lend ETH
- To mitigate counterparty risk and protect the lender, loans must be fully collateralized, and the collateral is locked in a smart contract
- Matching lenders with borrowers can be done in a variety of ways
 - The broad categories are P2P and pooled matching

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Decentralized Finance - Decentralized Lending Platforms

Collateralized Debt Markets

- P2P matching means that the person who is providing the liquidity lends the cryptoassets to specific borrowers
 - Consequently, the lender will only start to earn interest once there is a match
 - The advantage of this approach is that the parties agree on a time period and operate with fixed interest rates

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Decentralized Finance - Decentralized Lending Platforms

Collateralized Debt Markets

- Pooled loans use variable interest rates that are subject to supply and demand
 - The funds of all borrowers are aggregated in a single, smart contract-based lending pool, and lenders start to earn interest right when they deposit their funds in the pool
 - However, the interest rates are a function of the pool's utilization rate
 - When liquidity is readily available, loans will be cheap
 - When it is in great demand, loans will become more expensive
 - Lending pools have the additional advantage that they can perform maturity and size transformation while maintaining relatively high liquidity for the individual lender
- There is a large variety of lending protocols, some of the most popular ones are Aave and Compound

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Decentralized Finance - Decentralized Lending Platforms

Collateralized Debt Markets: Compound

- cTokens (e.g. cETH, cUSDC) are ERC-20 tokens minted by Compound when a user supplies assets to the protocol
 - A proportional claim on the underlying asset pool
 - The right to redeem underlying principal plus accrued interest
- How interest works
 - The number of cTokens you hold stays constant
 - The exchange rate between cTokens and the underlying asset increases over time
- Example
 - Deposit: 1,000 USDC
 - Receive: 50,000 cUSDC
- Initial exchange rate:
 - 1 cUSDC = 0.02 USDC
- After interest accrues:
 - Exchange rate becomes:
 - 1 cUSDC = 0.021 USDC
 - Redeeming 50,000 cUSDC → 1,050 USDC
- Interest is realized only at redemption, via a changing exchange rate

Decentralized Finance - Decentralized Lending Platforms

- You can also become a borrower by withdrawing assets deposited by others
- Borrower deposits 10 ETH
 - Receives cETH
 - Earns 2% APY on ETH
- Borrower borrows 12,000 USDC
 - Pays 6% APY on USDC
 - ETH remains locked as collateral

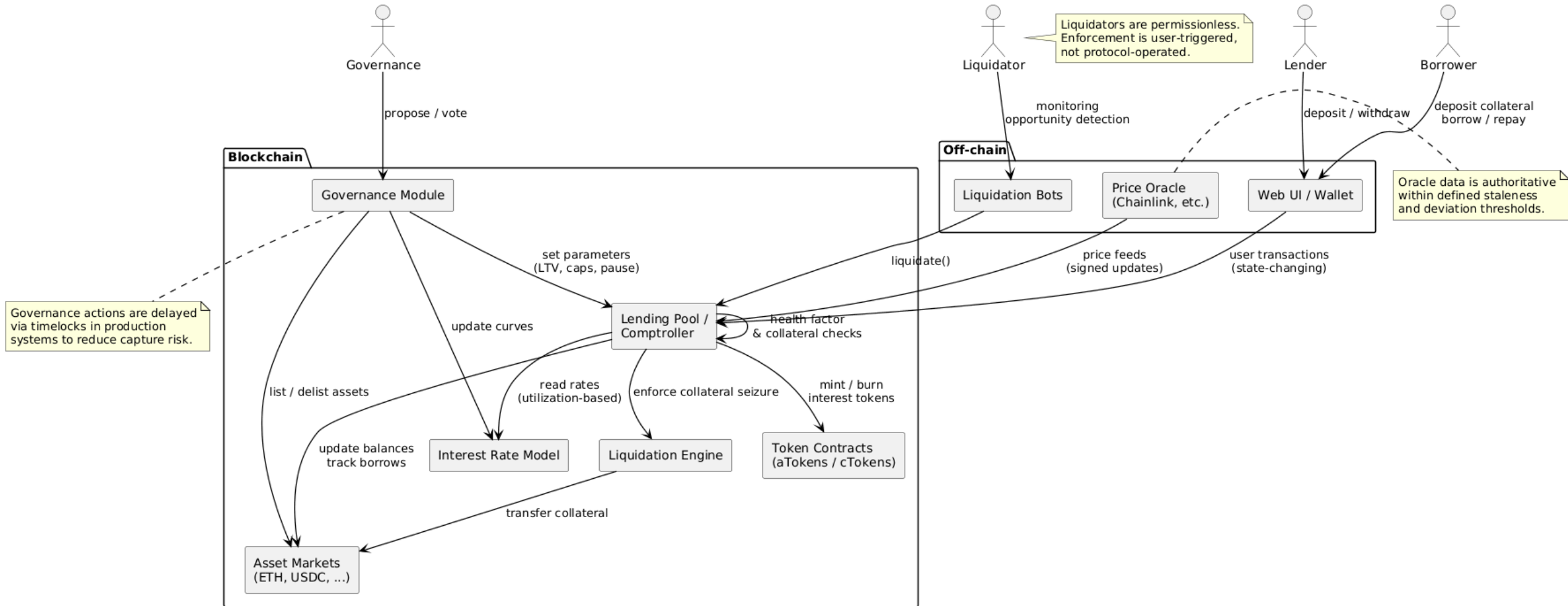
Over 1 year (simplified)

- Deposit side
 - $10 \text{ ETH} \times 2\% = 0.2 \text{ ETH}$ earned
- Borrow side
 - $\$12,000 \times 6\% = \720 interest paid
- The borrower is:
 - Earning ETH
 - Paying USDC
 - Exposed to ETH price risk
 - Exposed to liquidation risk



Decentralized Finance - Decentralized Lending Platforms

Decentralized Debt Market - System Architecture & Interactions



Decentralized Finance - Decentralized Derivatives

- Decentralized derivatives are tokens that derive their value from an underlying asset's performance, the outcome of an event, or the development of any other observable variable
- They usually require an oracle to track these variables and therefore introduce some dependencies and centralized components
- The dependencies can be reduced when the derivative contract uses multiple independent data sources

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Decentralized Finance - Decentralized Derivatives

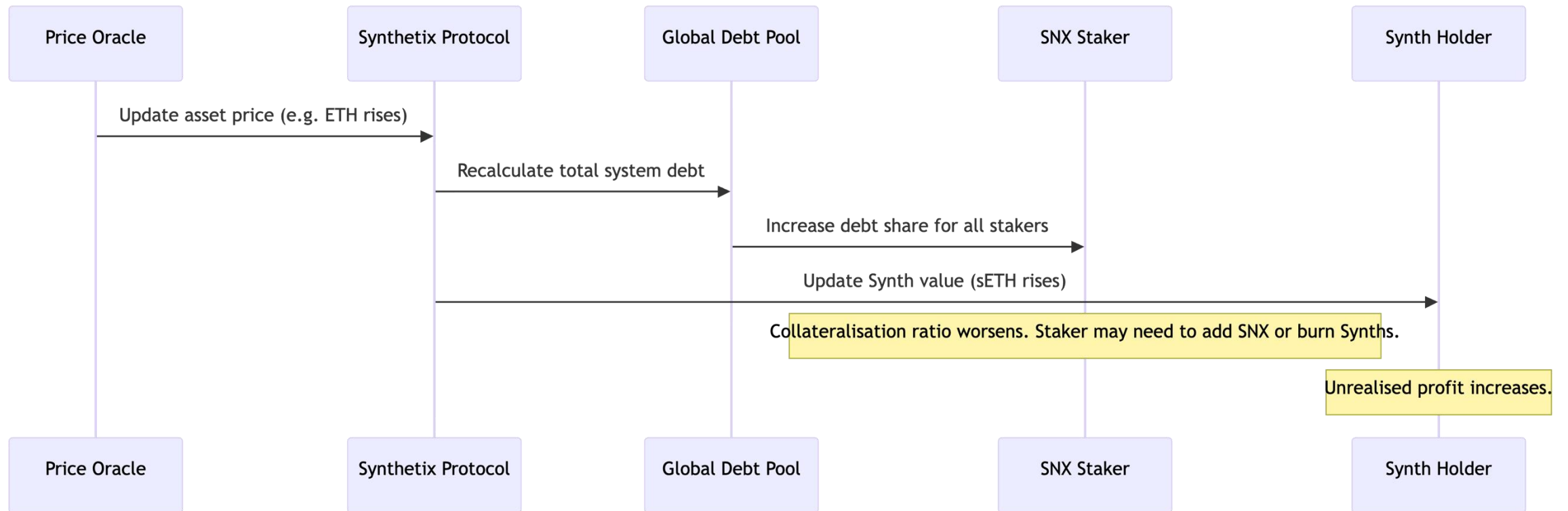
Asset-Based Derivative Tokens

- A popular derivative token platform is called Synthetix
- It is implemented so that the total debt pool of all participants increases or decreases depending on the aggregate price of all outstanding synthetic assets
- This ensures that tokens with the same underlying assets remain fungible; that is, redemption does not depend on the issuer
- The flip side of this design is that users assume additional risk when they mint assets, as their debt position will also be affected by everyone else's asset allocation

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Decentralized Finance - Decentralized Derivatives

- **Asset-Based Derivative Tokens**
- When Synth price goes up



- SNX Stakers are usually different from Synth traders

Decentralized Finance - Decentralized Derivatives

Event-Based Derivative Tokens

- Event-based derivative tokens can be based on any objectively observable variable with a known set of potential outcomes, a specified observation time, and a resolution source
 - Anyone can buy a full set of sub-tokens for a given event by locking 1 ETH in a smart contract
 - A complete set of sub-tokens consists of 1 sub-token for each potential outcome
 - These sub-tokens can be traded individually. When the market resolves, the smart contract's cryptoassets will be split among the sub-token owners of the winning outcome
 - In the absence of market distortions, each sub-token's ETH price should, therefore, correspond to the probability of the underlying outcome

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Decentralized Finance - Decentralized Derivatives

Event-Based Derivative Tokens

- Under certain circumstances, these prediction markets may serve as decentralized oracles for the likelihood of a future outcome
- Potential attack vectors include flawed or misleading question specifications, incomplete outcome sets that may render the event unresolvable, and the choice of unreliable or fraudulent resolution sources.
- One early protocol is Augur
 - It uses a multistage resolution and disputing process that should minimize the dependency on a single reporting source as much as possible
 - If the token holders do not agree with the designated reporter, they may start a dispute, which should eventually lead to the correct outcome

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Decentralized Finance - Decentralized Derivatives

‘Hairdryer used to trick weather sensor’ to win \$34,000 bet

Polymarket gamblers appear to net winnings as temperature readings at Charles de Gaulle airport spike unexpectedly



James Titcomb
Technology Editor

James Titcomb is The Telegraph's Technology Editor and has covered the tech industry for a decade from Silicon Valley and London. [See more](#) ✓

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Polymarket ran a daily market on the peak temperature in Paris, using the Charles de Gaulle airport weather station Credit: Peter Barritt/Alamy



Decentralized Finance - On-Chain Asset Management

- On-chain funds are mainly used for portfolio diversification
 - They allow users to invest in a basket of cryptoassets and employ a variety of strategies without having to handle the tokens individually
 - In contrast to traditional funds, the on-chain variant does not require a custodian
 - Instead, the cryptoassets are locked up in a smart contract
 - The investors never lose control over their funds, can withdraw or liquidate them, and can observe the smart contracts' token balances at any point in time

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Decentralized Finance - On-Chain Asset Management

- The smart contracts are set up in such a way that they follow a variety of simple strategies, including semi-automatic rebalancing of portfolio weights and trend trading, using moving averages.
- Alternatively, one or multiple fund managers can be selected to manage the fund actively
- In this case, the smart contract ensures that asset managers adhere to the predefined strategy and act in the investors' best interest
- Whenever someone invests in an on-chain fund, the corresponding smart contract issues fund tokens and transfers them to the investor's account
- These tokens represent partial ownership of the fund and allow token holders to redeem or liquidate their share of the assets

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Decentralized Finance - On-Chain Asset Management

- There are several implementations of on-chain fund protocols, including the Set Protocol, Enzyme Finance, Yearn Vaults, and Betoken
- All of these implementations are limited to ERC-20 tokens and Ether
- Moreover, they heavily depend on price oracles and third-party protocols, mainly for lending, trading, and the inclusion of low-volatility reference assets such as the Dai or USDC stablecoins

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Yield Farming

- Yield farming represents an immensely popular asset management activity in decentralized finance (DeFi)
- It involves supplying, borrowing, or staking crypto assets to earn an income in forms of transaction fees, interest, or participation rewards at different DeFi marketplaces

Xu, J., & Feng, Y. (2022). Reap the harvest on blockchain: A survey of yield farming protocols. *IEEE Transactions on Network and Service Management*, 20(1), 858-869.



Yield Farming

- Yield aggregators
 - Represented by Yearn, Beefy and Badger
 - The most classic and commonly known yield farming protocols are yield aggregators
 - In return for deposit into a yield farming pool, pool tokens that represent a fraction of the pool wealth are issued
 - Typically, the value of a pool token varies according to the total pool wealth

Xu, J., & Feng, Y. (2022). Reap the harvest on blockchain: A survey of yield farming protocols. *IEEE Transactions on Network and Service Management*, 20(1), 858-869.



Yield Farming

- Yield-bearing stablecoins
 - A yield-bearing stablecoin protocol works similar to a savings account with a bank
 - Instead of minting pool tokens, the protocol issues stablecoins to users as a form of certificate of deposit
 - The yield-generating nature of the protocol is reflected in the increase in the quantity of the stablecoins to their holders, as opposed to the value of the stablecoin token
 - The value is designed to remain stable to the peg

Xu, J., & Feng, Y. (2022). Reap the harvest on blockchain: A survey of yield farming protocols. *IEEE Transactions on Network and Service Management*, 20(1), 858-869.



Yield Farming

- Lottery protocols
 - A lottery protocol collect users' funds and issue them each a lottery ticket token in return
 - The protocol then performs yield farming under the hood
 - Instead of distributing yield proportionate to users' deposit, the protocol every once in a while randomly selects one or more winners who can pocket the yield of all participants
 - PoolTogether is one of the most popular protocols of this type

Xu, J., & Feng, Y. (2022). Reap the harvest on blockchain: A survey of yield farming protocols. *IEEE Transactions on Network and Service Management*, 20(1), 858-869.



Yield Farming

- NFT farming
 - With the popularity of nonfungible tokens (NFTs), groups began to explore involving NFTs in yield farming
 - The main goal of NFT farming is to create liquidity and utility for NFTs, especially in gaming space, thereby earning yields for token owners
 - Axie Infinity, ZooKeeper, Pulsar Farm, and MOBOX are typical platforms that provide NFT farming services

Xu, J., & Feng, Y. (2022). Reap the harvest on blockchain: A survey of yield farming protocols. *IEEE Transactions on Network and Service Management*, 20(1), 858-869.



Yield Farming

Sources of yield: Supply interest

- The most straightforward type of yield originates from lending
- In a bullish market, speculators are keen to borrow funds despite a high interest rate, in expectation of an appreciation in the assets of their leveraged long position
- A borrower wishing to increase their exposure to ETH, for example, may use ETH as collateral to borrow USDC, then repetitively exchanging USDC for ETH to deposit it as collateral to borrow more USDC, forming a “leveraging spiral”
- Compound and Aave, two major DeFi lending protocols while writing this paper have witnessed the borrow APY of USDC rising from 2-3% in May 2020 to as high as 10% in April 2021
- This specific kind of yield is incorporated in interest-bearing tokens, such as cTokens from Compound or aTokens from Aave

Xu, J., & Feng, Y. (2022). Reap the harvest on blockchain: A survey of yield farming protocols. *IEEE Transactions on Network and Service Management*, 20(1), 858-869.

Yield Farming

Sources of yield: Swap fee income

- Some tokens entitle users to part of the revenue that is going through the protocol
- These can be governance tokens or other kinds of tokens
- One example is the liquidity provider tokens in AMM-based DEXs
- By supplying liquidity into an AMM pool, users receive the fees that are paid by traders within that pool
- The higher the volume in that pool, the more fees that are generated, and the more a liquidity provider profits from this
- In Uniswap, a 0.3% fee is charged for every trade within a pool and goes fully to Liquidity Providers

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Yield Farming

Sources of yield: Participation reward

- Another yield source comes from liquidity mining programs, where early participants receive native tokens representing protocol ownership
- This incentivizes people to contribute funds into the protocol, and enhances decentralization as the protocol ownership is distributed to users
- The native tokens often have a governance functionality attached to them which is deemed valuable, as the token holders have a say in the future strategic direction of the project
- Native tokens sometimes also entitle holders to a share of the protocol revenue
- Another kind of revenue-sharing token, where users have to actively stake their tokens to receive a share of the revenue
 - For example, SUSHI holders that stake their SUSHI will get xSUSHI in return, which represents the proportional share of a pool that captures 0.05% of all trades on Sushiswap

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Yield Farming: Security Risks

- Flash loan attack
 - Flash loan attacks abuse the mechanism of flash loan protocols in which an attacker borrows a great deal of funds that do not require collateral
 - The attacker then manipulates the price of an asset in a very short period and quickly resells it to earn profits
 - Pancake Bunny Finance lost around 690,000 BUNNY tokens due to removal of liquidity and price manipulation by flash loan attacks
- Rug pull
 - A rug pull refers to the abandonment of a project by the project administrator after collecting investor's funds, leaving investors with valueless assets
 - Arbix Finance, a typical rug pull, drained around 10 m USD in users' assets directly from the vaults

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Yield Farming: Security Risks

- Reentrancy attack
 - Reentrancy attack occurs when a smart contract makes an external call to another smart contract
 - Then the another contract makes recursive calls back to the original function, intentionally or unintentionally withdraw funds
 - When the original contract fails to update its state before sending funds, the attacker can exploit this vulnerability to continuously drain the contract's funds
 - In September 2021, DAO Maker, was hacked for almost 4 million USD due to insecure smart contracts

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Yield Farming: Security Risks

- Key exploit
 - Due to poor access control of some DeFi systems, yield aggregators can be attacked by exploiting various keys (e.g., API key, wallet key) to tamper with the smart contracts or drain funds
 - For example, the Bent Finance utilized non-multisig wallets to deploy their project's smart contracts
 - Anyone who knows the appropriate private key can perform updates to the contracts, allowing attackers to inject malicious code and create the backdoor
 - In December 2021, an attacker leveraged this feature to drain 1.75 million USD worth of tokens from the pool

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Quiz

- Covers all 3 lectures
- 90 minutes
 - Quiz will start at 6:45pm
- 20 multiple choice (10%)
- 2 short answer questions (30%)
 - Each question may have parts
- NTULearn
 - Respondus Lockdown Browser is required
 - You must be physically present in the lecture room to take the quiz

